Executive Summary

This report analyzes Reddit technology and programming comments using network analysis and natural language processing techniques to understand how users interact and what topics and sentiments appear within discussion communities. The dataset was collected from selected technology and programming-related subreddits using the PullPush Reddit API. After preprocessing, the final dataset contained 5,437 usable comments across 28 subreddits.

A directed weighted user reply network was constructed where nodes represent Reddit users and edges represent reply interactions between users. The final network contained 1,692 users and 1,991 reply relationships. Centrality measures, including in-degree, out-degree, weighted degree, PageRank, and betweenness centrality, were used to identify influential, active, and bridge users. Louvain community detection was then applied to identify interaction communities within the network, resulting in 278 detected communities with a modularity score of 0.953.

In addition to network analysis, the report applies NLP techniques to examine comment content. Sentiment analysis was performed using both VADER and RoBERTa, with RoBERTa selected as the main sentiment model due to its stronger contextual understanding. RoBERTa classified most comments as neutral, which reflects the explanatory and technical nature of programming-related discussions. Topic modeling using LDA identified major discussion themes related to AI, programming, Linux, data, and technical problem-solving.

Overall, the analysis shows that combining network analysis with NLP provides a deeper understanding of Reddit technology communities by revealing not only who interacts with whom, but also what users discuss and how sentiment varies across communities.

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1. Introduction

1.1 Background

Reddit is a discussion-based social media platform where users participate in topic-specific communities called subreddits. Its threaded comment structure makes it suitable for both social network analysis and text-based analysis because users can post content, reply to others, ask questions, share opinions, and discuss technical issues. Technology and programming-related subreddits are especially relevant because they contain active discussions about software development, programming languages, artificial intelligence, data science, cybersecurity, operating systems, and broader technology topics. In this project, user reply interactions can be represented as a network, where users are nodes and replies between users are edges. At the same time, Reddit comments provide rich textual data for NLP analysis. By applying sentiment analysis, keyword analysis, and topic modeling, the report can identify discussion themes and emotional tone. Therefore, combining network analysis with NLP provides a more complete understanding of Reddit technology and programming communities.

1.2 Project Aim

This report analyzes Reddit technology and programming comments using network analysis and NLP/text analysis to understand how users interact and what topics and sentiments appear in those interaction communities.

1.3 Research Questions

The main research questions for this project are:

1. Which users are most central in the Reddit reply network?
2. Which users act as bridges between different discussion groups?
3. What communities are detected in the user reply network?
4. What topics and sentiments appear in Reddit technology/programming discussions?
5. How do LDA topics and sentiment patterns connect with Louvain network communities?

1.4 Analysis Overview

This report follows a structured analytical process. First, Reddit comments were collected from selected technology and programming-related subreddits. The dataset was then cleaned by removing duplicate, deleted, and removed comments. After preprocessing, a directed weighted user reply network was created using comment-to-comment reply relationships. Network centrality measures were used to identify important users, while Louvain community detection was applied to identify user interaction groups.

After the network analysis, NLP techniques were used to analyze the content of Reddit comments. Sentiment analysis was performed using VADER and RoBERTa, and topic modeling was conducted using Latent Dirichlet Allocation. Finally, the detected Louvain communities were connected with sentiment and topic patterns to understand how different interaction groups discuss different themes.

Research Question	Method Used	Main Output
Which users are most central in the Reddit reply network?	In-degree, out-degree, weighted degree, PageRank	Identification of central and influential users
Which users act as bridges between different discussion groups?	Betweenness centrality	Identification of bridge users

What communities are detected in the user reply network?	Louvain community detection	User interaction communities
What topics and sentiments appear in Reddit technology/programming discussions?	RoBERTa, VADER, TF-IDF, LDA	Sentiment and topic patterns
How do LDA topics and sentiment patterns connect with Louvain network communities?	Community-level sentiment and topic analysis	Interpretation of topics and sentiment within detected communities

Table 1: Research Questions and Analysis Methods

2. Data Collection and Dataset Description

2.1 Data Source

The dataset used in this report was collected from Reddit, a discussion-based social media platform where users participate in topic-specific communities known as subreddits. Reddit was selected because its threaded comment structure supports both text analysis and network analysis. The comment text can be used for sentiment analysis, keyword analysis, and topic modeling, while reply relationships between comments can be used to construct a user-to-user interaction network.

The data was collected from technology and programming-related subreddits using the PullPush Reddit API. The collection process was implemented in the notebook fetch_selected_reddit_comments.ipynb. The final collected dataset was exported as reddit_technology_programming_comments.csv and used as the input dataset for the main analysis notebook, technology_and_programming_communities_reddit_analysis.

2.2 Data Collection Process

The data collection process had two stages. First, Reddit submissions were collected from selected technology and programming-related subreddits. Second, comments were collected from those submissions. The dataset captured key fields such as subreddit, post ID, post title, comment ID, author, comment text, parent ID, score, and timestamp. The parent_id field was especially important because it identified whether comments replied to posts or other comments, supporting reply network construction.

2.3 Selected Reddit Communities

The dataset was collected from 28 Reddit communities related to technology, programming, artificial intelligence, data science, cybersecurity, software development, and automation. The selected subreddits were: *programming, learnprogramming, Python, java, javascript, learnpython, webdev, coding, technology, androiddev, GenAI4all, generativeAI, OpenAI, linux, C_Programming, CyberCrime, cybersecurity, datascience, dataanalysis, dataengineering, algorithms, analytics, MachineLearning, learnmachinelearning, deeplearning, automation, aiengineering, and neuralnetworks.*

These subreddits were selected because they represent different areas of technology and programming discussion.

This selection provides a broad dataset for analyzing both user interaction patterns and discussion content across Reddit technology and programming communities. Instead of focusing on only one technical subreddit, the dataset captures different types of discussions, which supports the aim of understanding how users interact and what topics and sentiments appear across multiple technology-related communities.

2.4 Raw Dataset Summary

Before preprocessing, the collected dataset contained 6,225 Reddit comments from 28 subreddits, covering 680 Reddit posts and 3,211 unique authors. The dataset included 6,123 unique comment IDs, meaning 102 duplicate comment IDs were identified. Although there were no missing values in the main collected fields, some duplicate, deleted, and removed records were found. These records were handled during preprocessing to improve data quality before conducting the network analysis and NLP analysis. Overall, the raw dataset was large enough to support both user interaction analysis and text-based analysis, but cleaning was required before creating the final analytical dataset.

2.5 Dataset Variables

The dataset contains variables required for both network analysis and NLP/text analysis. The interaction analysis mainly depends on the author, comment_id, and parent_id fields, while the text analysis mainly depends on the body, subreddit, and timestamp fields.

Variable	Description	Main Use in Analysis
subreddit	Name of the Reddit community where the comment was posted	Subreddit-level comparison
post_id	Unique identifier of the Reddit post	Linking comments to posts
post_title	Title of the Reddit post	Contextual understanding of discussion topics
comment_id	Unique identifier of each Reddit comment	Duplicate removal and reply matching
author	Reddit username of the comment author	User node creation
body	Text content of the Reddit comment	Sentiment analysis and topic modeling
parent_id	Identifier of the parent post or parent comment	Reply network construction
link_id	Identifier linking the comment to the original post	Post-level grouping
score	Reddit score of the comment	Descriptive analysis
created_utc	Comment creation time in Unix timestamp format	Time conversion
created_datetime	Converted readable timestamp	Time-based exploration

Table 2: Main Dataset Variables

The comment_id, author, and parent_id fields were essential for building the reply network because they allowed the analysis to identify direct reply relationships between users. The body field was essential for the NLP component because it contained the text used for sentiment analysis, TF-IDF keyword analysis, and LDA topic modeling.

2.6 Relevance of the Dataset to the Research Questions

The dataset is suitable for the research questions because it contains both interaction data and textual data. The reply-related fields allow Reddit users to be modeled as nodes and replies between users to be modeled as directed edges. This supports the identification of central users, bridge users, and interaction communities within the Reddit reply network.

At the same time, the comment text allows the analysis of sentiment and discussion topics across technology and programming communities. This makes it possible to examine not only how users interact, but also what they discuss and how the tone of discussion varies across different subreddits and detected Louvain communities. Therefore, the dataset directly supports the combined network analysis and NLP-based approach used in this report.

3. Data Preprocessing

Data preprocessing was performed to improve the quality and reliability of the Reddit dataset before conducting network analysis and NLP/text analysis. The raw dataset contained useful comment and reply information, but it also included duplicate records, deleted authors, removed comments, and timestamp fields that required conversion. These issues were addressed to ensure that the final analysis was based on valid and meaningful Reddit comments.

3.1 Data Quality Checks

The first preprocessing step checked the raw dataset for missing values, duplicate comment IDs, deleted authors, removed comments, and invalid text. No missing values were found in the main collected fields, but duplicate comment IDs were identified and removed so that each Reddit comment appeared only once. The detailed missing value check is provided in Appendix Table A1, and the duplicate comment ID check is provided in Appendix Table A2. Deleted authors and deleted or removed comments were also excluded because reliable usernames and meaningful text were required for network analysis, sentiment analysis, and topic modeling. See Appendix Table A3.

3.2 Timestamp Conversion

The created_utc field was converted from Unix timestamp format into a readable datetime format. This conversion allowed the dataset to be interpreted more clearly and supported time-based exploration of Reddit comment activity. The converted timestamp field was used to identify the date range of the dataset and to support exploratory analysis such as comment activity by hour of day.

3.3 Reply Target Identification

A key preprocessing step was identifying whether each comment replied to a post or another comment using the parent_id field. In Reddit data, parent_id values beginning with t3_ indicate replies to posts, while values beginning with t1_ indicate replies to comments. This distinction was essential for constructing the user reply network. Only comment-to-comment replies were used to create direct user-to-user edges because they identify both the replying user and the target user. Post replies were retained for general dataset and NLP analysis but excluded from user-to-user network edges. A detailed breakdown is provided in Appendix Table A4

3.4 Text Preparation for NLP Analysis

The comment text was prepared for NLP analysis by creating cleaned text fields. This helped reduce noise before applying sentiment analysis, TF-IDF keyword extraction, and LDA topic modeling. Reddit comments often contain informal language, URLs, symbols, abbreviations, and technical expressions. Therefore, the cleaning process was applied carefully so that useful technology and programming-related terms were retained while unnecessary noise was reduced.

This step was important because the quality of text preprocessing directly affects the reliability of sentiment classification and topic modeling. Cleaned text allowed the NLP models to focus more effectively on meaningful terms and discussion themes.

3.5 Final Cleaned Dataset

After preprocessing, the final cleaned dataset contained 5,437 usable Reddit comments across 28 subreddits. Duplicate comments, deleted authors, and removed or deleted comments were excluded, leaving 0 duplicate comment IDs and 0 missing values in the final dataset. The cleaned data included

3,103 unique authors and 549 unique posts, covering the period from 4 March 2025 to 19 May 2025. Of the cleaned comments, 3,056 were replies to other comments, while 2,381 were replies directly to posts. This final dataset provided a reliable basis for both network analysis and NLP/text analysis.

4. Exploratory Data Analysis

Exploratory data analysis was conducted to understand the structure and distribution of the cleaned Reddit dataset before applying network analysis and NLP techniques. This step helped identify which subreddits contributed the most comments, how many users participated in each community, and whether discussion activity was evenly distributed across the selected technology and programming communities.

4.1 Subreddit-Level Comment Distribution

The cleaned dataset contained 5,437 comments across 28 subreddits. However, the distribution of comments was not equal across all communities. Some subreddits contributed a much larger number of comments, while others had relatively small amounts of discussion activity. This is important because subreddits with more comments may have a stronger influence on the overall sentiment and topic modeling results.

Table 3 shows the top 5 subreddits by number of comments. The full subreddit-level summary for all 28 subreddits is provided in Appendix Table A5.

Subreddit	Number of Comments	Number of Posts	Number of Authors	Average Score	Average Word Count
GenAI4all	863	40	534	1.03	32.29
datascience	772	29	465	1.06	47.85
java	502	29	256	1.83	48.69
technology	475	20	321	6.63	39.10
linux	454	18	282	1.00	40.47

Table 3: Top 5 Subreddits by Comment Volume

The subreddit-level summary shows that GenAI4all contributed the highest number of comments, with 863 comments from 534 authors. This was followed by datascience with 772 comments and java with 502 comments. These results show that artificial intelligence, data science, and programming language discussions were strongly represented in the dataset.

The technology subreddit had 475 comments, which was fewer than GenAI4all, datascience, and java, but it had the highest average score among the top 10 subreddits. This suggests that comments in technology received comparatively stronger engagement. In contrast, subreddits such as automation, programming, and learnpython contributed fewer comments but still provided useful discussion diversity across technical topics.

4.2 Visual Distribution of Comments by Subreddit

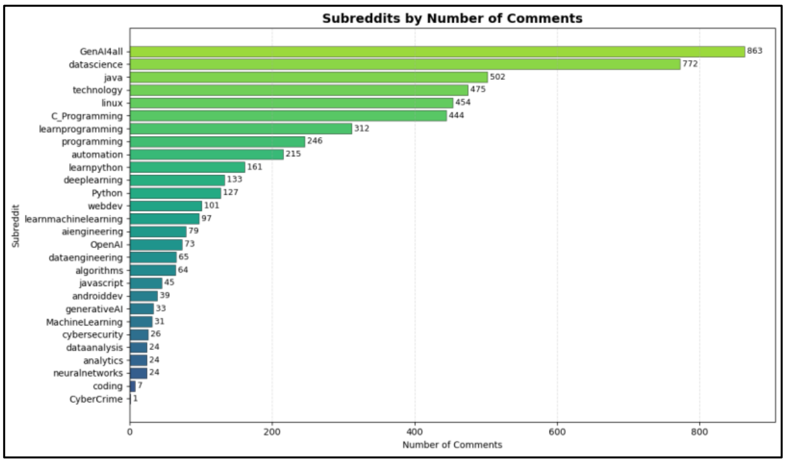

Figure 1: Subreddits by Number of Comments

Figure 1 visualizes the number of comments contributed by each subreddit. The figure confirms that the dataset is concentrated around a small number of highly active communities. GenAI4all and datascience are the two largest contributors, while communities such as CyberCrime, coding, dataanalysis, analytics, and neuralnetworks contributed far fewer comments.

This imbalance is important for interpretation because larger communities may have more influence on overall topic and sentiment patterns. However, smaller subreddits still provide useful context because they add diversity to the dataset and represent more specialized technical discussions.

4.3 Comment Activity by Hour of Day

The timestamp field was used to examine when Reddit comments appeared in the dataset. This analysis provides a descriptive view of comment activity over time and helps identify whether discussion activity was concentrated during particular hours.

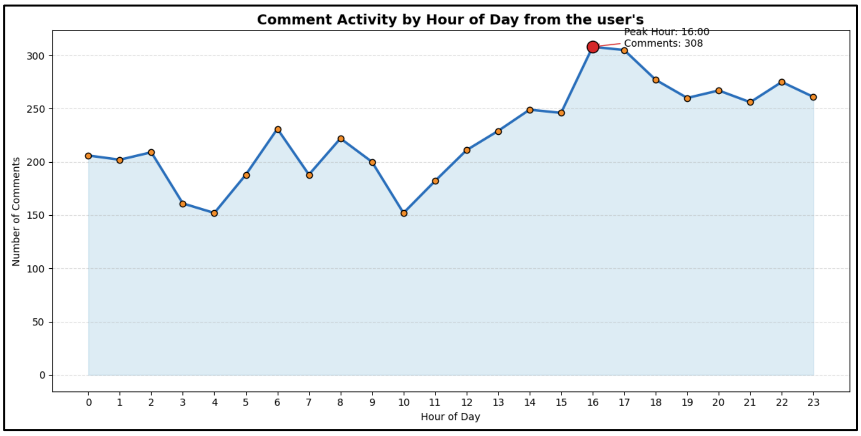

Figure 2: Comment Activity by Hour of Day

Figure 2 shows the distribution of comments by hour of day. This figure was used as an exploratory check rather than a main research result. The main purpose was to understand the temporal structure of the dataset and confirm that the collected comments were distributed across different times rather than being concentrated in only one short period.

Although time-based activity is not the main focus of this report, it supports the overall understanding of the dataset before conducting network analysis and NLP analysis. The later sections focus more directly on user interaction patterns, centrality, community detection, sentiment, and topic modeling.

4.4 Summary of Exploratory Findings

The exploratory analysis shows that the dataset covers a broad range of technology and programming discussions, with strong representation from AI, data science, Java, Linux, and general technology communities. However, comment activity was uneven across subreddits, so larger communities may influence later sentiment and topic modeling results. Overall, the cleaned dataset contains enough user and comment activity to support reply network construction, centrality analysis, community detection, sentiment analysis, and topic modeling.

5. User Reply Network Construction

A user reply network was constructed to analyze how Reddit users interacted with each other across technology and programming discussions. In this network, each node represents a Reddit user, and each directed edge represents a reply from one user to another user. This approach allows the analysis to move beyond simple comment counts and examine the structure of user interactions within Reddit discussions.

The network was created using comment-to-comment reply relationships. Comments that replied directly to posts were retained for general dataset analysis and NLP/text analysis, but they were not used to create user-to-user edges because they did not identify another comment author as the direct target user. Therefore, only replies from one comment to another comment were used to construct the final user reply network.

5.1 Network Definition

The Reddit reply network was designed as a directed weighted network. Direction was important because each reply has a clear source and target. For example, if User A replied to a comment written by User B, the edge was recorded as User A → User B. Edge weight was used to represent repeated reply interactions between the same pair of users.

Network Element	Definition
Node	A Reddit user/author
Directed edge	A reply from one user to another user
Edge source	The user who wrote the reply
Edge target	The user whose comment received the reply
Edge direction	Source user → target user
Edge weight	Number of replies from the same source user to the same target user
Network type	Directed weighted user reply network
Exported file	user_reply_network.graphml

Table 4: Network Construction Design

This network design was appropriate for the research questions because it allowed the identification of central users, active users, bridge users, and interaction communities. A directed weighted structure was also more informative than an undirected network because it preserved both the direction and frequency of user reply relationships.

5.2 Reply Relationship Extraction

The parent_id field was used to identify comment-to-comment replies. In Reddit data, a parent_id beginning with t1_ represents a reply to another comment, while a parent_id beginning with t3_ represents a reply to a post. For network construction, only t1_ comment replies were used because these records could be matched to another comment and therefore to another user.

The reply extraction process involved matching each replying comment with its parent comment. Once the parent comment was identified, the author of the replying comment became the source user, and the author of the parent comment became the target user. This created a direct user-to-user relationship.

The construction process followed these steps:

1. Filter comments where the parent ID represented another comment.
2. Match the parent comment ID with the original comment record.
3. Identify the replying user as the source node.
4. Identify the parent comment author as the target node.

5. Remove invalid reply records where required.
6. Aggregate repeated replies between the same source and target users into weighted edges.

Sample outputs from the reply extraction process are provided in Appendix Table A6, while Appendix Table A7 shows the weighted edge list created after aggregating repeated user-to-user replies. Appendix Table A8 provides sample node-level attributes used to describe user activity in the final network.

5.3 Weighted Edge Creation

Repeated replies between the same source and target users were aggregated into weighted directed edges. This means that if one user replied to another user multiple times, the relationship was represented as one edge with a higher weight rather than several separate edges. This approach captured interaction strength, not only whether two users interacted. The weighted edge list was then used to construct the final Reddit user reply network, which was exported as user_reply_network.graphml for use in tools such as NetworkX or Gephi.

5.4 Network Summary and Relevance to Research Questions

The final user reply network contained 1,692 user nodes and 1,991 directed weighted reply edges, representing 2,366 aggregated reply interactions. The very low-density value of 0.000696 shows that the network was sparse, with most users interacting with only a small number of others. The network also contained 264 weakly connected components and 1,242 strongly connected components, indicating that Reddit discussions were fragmented across many smaller groups and sub-conversations rather than forming one unified network. This structure is typical of Reddit, where users usually interact within specific posts or threads. The constructed network directly supports the first three research questions by enabling centrality analysis to identify important users, betweenness centrality to identify bridge users, and Louvain community detection to identify interaction groups.

6. Network Analysis and Community Detection

Network analysis was performed on the directed weighted Reddit user reply network to identify important users, bridge users, and interaction communities. This section focuses on the structural patterns of user interaction rather than the textual content of comments.

The results in this section directly address the first three research questions: which users are most central in the Reddit reply network, which users act as bridges between different discussion groups, and what communities are detected in the user reply network.

6.1 Centrality Measures

Several centrality measures were calculated to identify different types of user roles within the Reddit reply network. Since the network was directed and weighted, the analysis considered both the direction of replies and the frequency of repeated interactions between users.

Centrality Measure	Meaning in This Project	Interpretation
In-degree	Number of users who replied to a user	Users who attracted replies from others
Out-degree	Number of users a user replied to	Users who actively participated in discussions
Weighted in-degree	Total number of replies received	Users who received repeated replies

Weighted out-degree	Total number of replies sent	Users who repeatedly replied to others
PageRank	Importance based on receiving replies from other important users	Structurally influential users
Betweenness centrality	Frequency with which a user lies on paths between other users	Bridge users connecting discussion groups

Table 5: Network Centrality Measures Used

These measures were used because user importance in a reply network can take different forms. A user may be important because they receive many replies, actively reply to others, receive replies from structurally important users, or connect otherwise separate discussion groups. Therefore, multiple centrality measures were required to provide a more complete interpretation of user roles in the Reddit reply network.

6.2 Identification of Central and Bridge Users

The centrality results showed that users played different roles in the Reddit reply network. Weighted in-degree identified users who received many replies and attracted discussion, while weighted out-degree identified users who actively replied to others. PageRank was used to identify structurally influential users by considering both incoming replies and the importance of users providing those replies. Betweenness centrality identified bridge users who connected different parts of the network, making them important for linking separate discussion groups or sub-conversations.

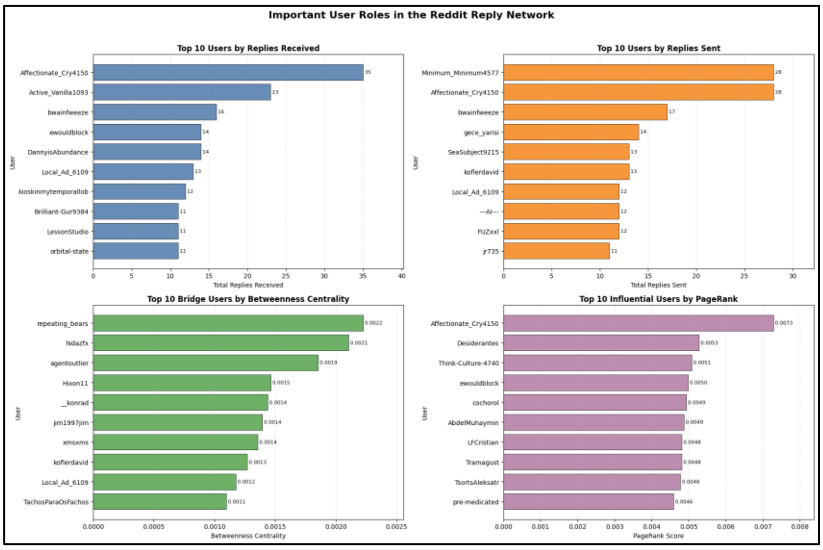

Figure 3: Important User Roles in the Reddit Reply Network

Figure 3 presents the top users across four network roles: most replied-to users, most active replying users, influential users by PageRank, and bridge users by betweenness centrality. The figure shows that the same user does not always dominate every centrality measure. For example, a user may receive many replies but may not necessarily act as a bridge between groups. Similarly, a bridge user may not have the highest number of total replies but may still be structurally important because they connect different parts of the network.

The detailed top 10 user rankings for weighted in-degree, weighted out-degree, PageRank, and betweenness centrality are provided in Appendix Table A9, Appendix Table A10, Appendix Table A11, and Appendix Table A12.

6.3 Louvain Community Detection

Louvain community detection was applied to identify groups of users with stronger internal connections in the Reddit reply network. The directed network was converted to an undirected version because Louvain detects communities based on connection strength and modularity. The method identified 278 communities, with a modularity score of 0.953, indicating very strong separation between interaction groups. The largest community contained 96 users, while the smallest communities contained 2 users. These results show that Reddit interactions were highly fragmented and organized into smaller discussion clusters rather than one unified network. However, because the network was sparse and Reddit discussions often occur within specific posts and nested comment threads, many detected communities likely represent thread-level or subreddit-specific interaction groups.

6.4 Interpretation of Major Louvain Communities

The largest Louvain communities were interpreted by examining their number of users, total comments, dominant subreddit, top users, and top keywords. This helped connect the structural communities to meaningful discussion contexts.

Community	Community Label	Users	Total Comments	Dominant Subreddit	Top Users	Top Keywords
184	programming interaction group	96	197	programming	pre-medicated, LessonStudio, Local_Ad_6109	100k, url, handle, urls, aws, database, code
89	java interaction group	87	238	java	ewouldblock, Hixon11, Diligent_End8130	java, code, python, string, optional, zgc, language

Table 6: Top Louvain Communities by Size and Theme

Table 6 shows that the largest Louvain communities were strongly associated with specific subreddits and technical themes. For example, Community 184 was mainly connected to the programming subreddit and included keywords related to URLs, AWS, databases, and code. Community 89 is connected to the java subreddit, but keyword patterns suggest different technical discussion focuses.

These results show that the Louvain communities are not only structural clusters but also meaningful discussion groups. The communities reflect the way Reddit users interact around specific technical topics, subreddits, and conversation threads.

6.5 Network Visualization

Network visualization was created to support the interpretation of the Reddit user reply network. The visualization shows a filtered version of the network containing the top 80 users based on PageRank.

To make the network easier to interpret, a second graph was created by filtering the network to the top 80 users by PageRank. In this filtered visualization, nodes represent Reddit users, edges represent reply relationships, node size represents PageRank, node color represents Louvain community membership, and edge thickness represents reply weight.

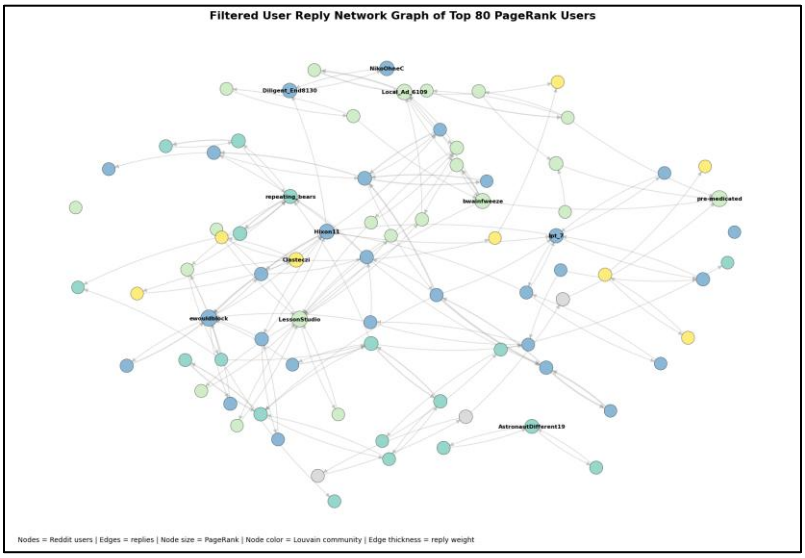

Figure 4: Filtered User Reply Network Graph of Top 80 PageRank Users

Figure 4 provides a clearer view of the most structurally important users in the reply network. The filtered graph shows visible clusters of users, which supports the Louvain community detection results. Some users appear in more central positions, while others are located within smaller local clusters. This reflects the nature of Reddit discussions, where users often interact within specific threads, subreddits, or topic-based discussion groups.

Figure 4 shows the overall structure and the interpretable core of the Reddit reply network. The full GraphML graph demonstrates the size and fragmentation of the complete network, while the filtered PageRank graph provides a clearer view of important users and community structure.

6.6 Summary of Network Analysis Findings

The network analysis shows that Reddit technology and programming discussions form a sparse but meaningful reply network. Centrality measures identified different user roles, including users who received many replies, users who actively replied to others, influential users, and bridge users. Louvain community detection identified 278 communities with a high modularity score of 0.953, suggesting strong separation between interaction groups.

Overall, this section shows that Reddit discussion structure is shaped by smaller user interaction clusters. These findings provide the network foundation for the next sections of the report, where sentiment analysis and topic modeling are used to understand what users discussed within these technology and programming communities.

7. NLP, Sentiment Analysis, and Topic Modeling

After constructing the Reddit user reply network, NLP techniques were applied to analyze the textual content of the comments. This section focuses on identifying sentiment patterns, important keywords, and major discussion topics across Reddit technology and programming communities. The analysis used VADER and RoBERTa for sentiment classification, TF-IDF for keyword analysis, and LDA for topic modeling. This section mainly addresses the fourth research question:

What topics and sentiments appear in Reddit technology/programming discussions?

7.1 NLP Dataset Preparation

The cleaned Reddit dataset used for NLP analysis contained 5,437 comments from 28 subreddits and 3,103 unique authors. Cleaned comment text was used for sentiment analysis, TF-IDF keyword extraction, and LDA topic modeling. The data was also merged with Louvain community assignments, with 3,764 comments linked to a detected community. Since community labels were only available for users in the reply network, general NLP analysis used all comments, while community-level analysis is discussed in Section 8.

7.2 Sentiment Analysis Using VADER and RoBERTa

Sentiment analysis was conducted using two models: VADER and RoBERTa. VADER was used as a baseline lexicon-based sentiment model. It is commonly used for social media text because it can classify sentiment based on word polarity, punctuation, and intensity. However, VADER may not always capture the meaning of technical, sarcastic, or context-dependent Reddit comments because it relies heavily on predefined sentiment rules.

RoBERTa was used as the main sentiment model for final interpretation. RoBERTa is a transformer-based model that considers the context of words in a sentence, making it more suitable for technical Reddit comments. This was important because programming and technology discussions often contain explanations, disagreement, troubleshooting, criticism, and neutral technical advice.

Model	Sentiment Label	Number of Comments	Percentage	Average Sentiment Score
VADER	Positive	2,998	55.141%	0.580
VADER	Neutral	1,195	21.979%	0.000
VADER	Negative	1,244	22.880%	-0.455
RoBERTa	Positive	1,031	18.963%	0.744
RoBERTa	Neutral	2,793	51.370%	-0.034
RoBERTa	Negative	1,613	29.667%	-0.679

Table 7: VADER and RoBERTa Sentiment Summary

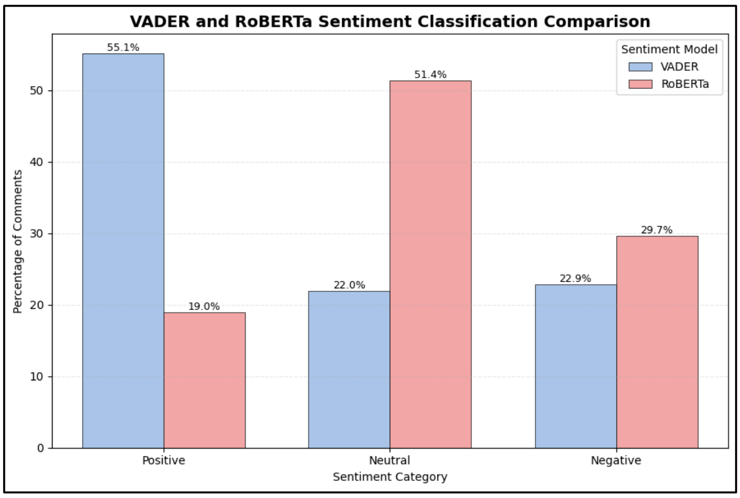

Figure 5: VADER and RoBERTa Sentiment Classification Comparison

Figure 5 shows that the two sentiment models produced substantially different results. VADER classified 55.1% of comments as positive, while RoBERTa classified only 19.0% as positive. In contrast, RoBERTa classified 51.4% of comments as neutral, compared with only 22.0% for VADER.

This difference is important because technical Reddit comments often contain words that may appear positive or negative in isolation, but the overall comment may be neutral, explanatory, or problem-focused. For example, programming comments may discuss errors, code improvements, performance, or software limitations without expressing strong emotional sentiment. For this reason, RoBERTa was used as the main model for the final sentiment interpretation.

7.3 Sentiment by Subreddit

After selecting RoBERTa as the main sentiment model, sentiment was analyzed across subreddits. The RoBERTa sentiment score was calculated as positive probability minus negative probability. Therefore, a score closer to +1 indicates more positive sentiment, a score close to 0 indicates neutral or mixed sentiment, and a score closer to -1 indicates more negative sentiment.

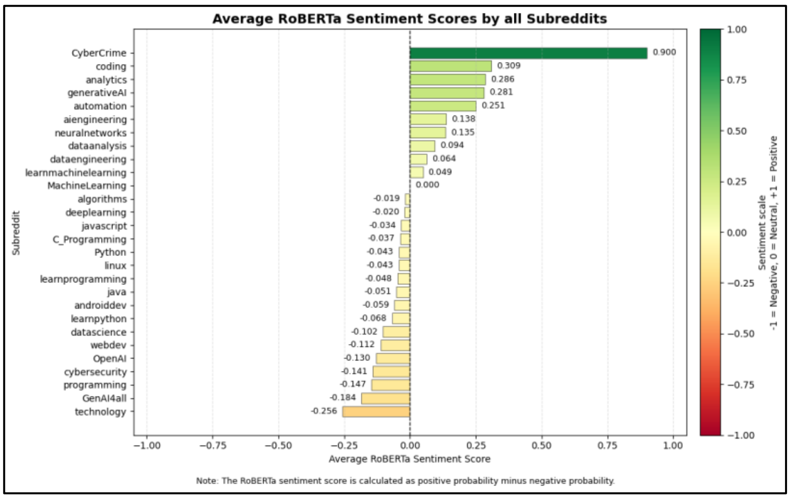

Figure 6: Average RoBERTa Sentiment Scores by All Subreddits

Figure 6 shows that average sentiment varied across the selected subreddits. Some subreddits, such as automation, generativeAI, analytics, and coding, showed more positive average sentiment. However, results for very small subreddits should be interpreted cautiously. For example, CyberCrime had the highest average sentiment score, but this was based on only one comment, so it should not be treated as a strong general pattern.

The most negative average sentiment appeared in subreddits such as technology, GenAI4all, programming, cybersecurity, OpenAI, and webdev. This does not necessarily mean that these communities were generally negative. In technical communities, negative sentiment may reflect criticism, disagreement, bug reports, troubleshooting, frustration with tools, or discussion of risks and limitations.

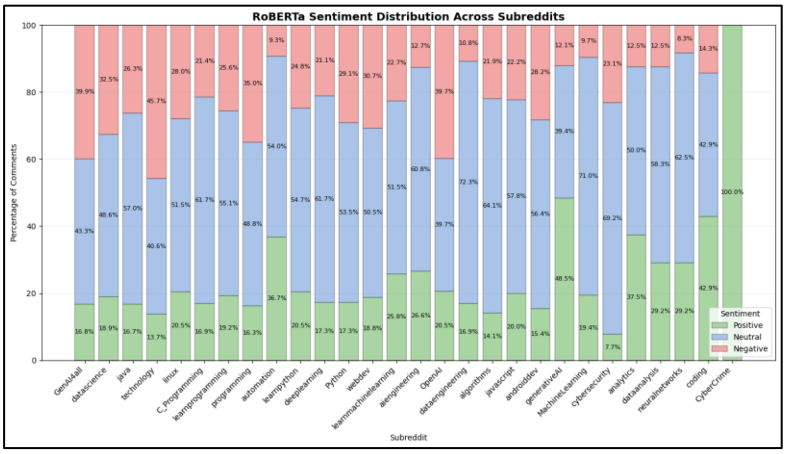

Figure 7: RoBERTa Sentiment Distribution Across Subreddits

Figure 7 provides a more detailed view of sentiment composition across subreddits. While Figure 6 shows average sentiment scores, Figure 7 shows the percentage of positive, neutral, and negative comments within each subreddit.

The distribution shows that neutral sentiment was common across most subreddits. This supports the interpretation that many Reddit technology and programming comments were informational, explanatory, or discussion-based rather than strongly emotional. Some subreddits had a larger proportion of negative comments, particularly technology, GenAI4all, OpenAI, and programming. These patterns may reflect more debate-oriented discussions, criticism, or problem-focused conversations.

The full subreddit-level RoBERTa sentiment summary is provided in Appendix Table A13.

7.4 TF-IDF Keyword Analysis

TF-IDF keyword analysis was used to identify important terms in the cleaned Reddit comments. Unlike simple word frequency, TF-IDF gives higher importance to terms that appear frequently in some comments but are not equally common across all comments. This makes it useful for identifying distinctive words within the dataset.

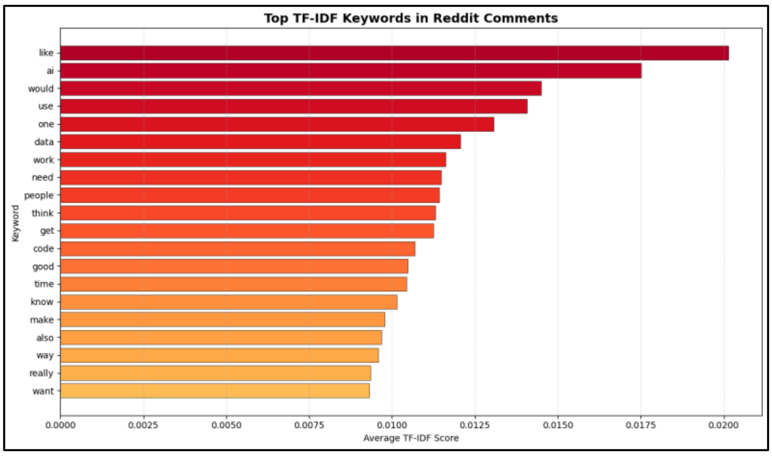

Figure 9: Top TF-IDF Keywords in Reddit Comments

Figure 9 shows the top TF-IDF keywords across the Reddit comments, including like, ai, would, use, data, work, people, code, and time. These terms indicate that the dataset contains both technical vocabulary and general discussion language. The keyword ai confirms strong artificial intelligence-related discussion, while data, code, and work reflect programming, data science, and software-related conversations. TF-IDF was used as a supporting analysis before topic modeling and confirmed the dataset's relevance to technology and programming discussions.

7.5 LDA Topic Modeling

Latent Dirichlet Allocation was used to identify major discussion themes in the cleaned Reddit comments. LDA is a topic modeling method that groups words into topics based on patterns of co-occurrence across documents. In this project, six topics were generated, and each comment was assigned a dominant topic.

The topics were interpreted using their most important keywords, number of comments, average topic probability, and sentiment distribution.

Topic	Topic Label	Number of Comments	Average Topic Probability	Average Sentiment	Positive %	Neutral %	Negative %
1	linux, java, windows	602	0.609	-0.023	18.605	56.645	24.751
2	data, use, would	855	0.676	-0.059	10.175	72.047	17.778
3	ai, work, time	1,265	0.685	-0.000	25.929	46.957	27.115
4	people, ai, would	1,107	0.693	-0.242	14.905	40.108	44.986
5	gt, lt, number	342	0.677	-0.124	11.696	61.404	26.901
6	like, ai, things	1,266	0.683	-0.038	23.618	46.445	29.937

Table 8: LDA Topic Summary

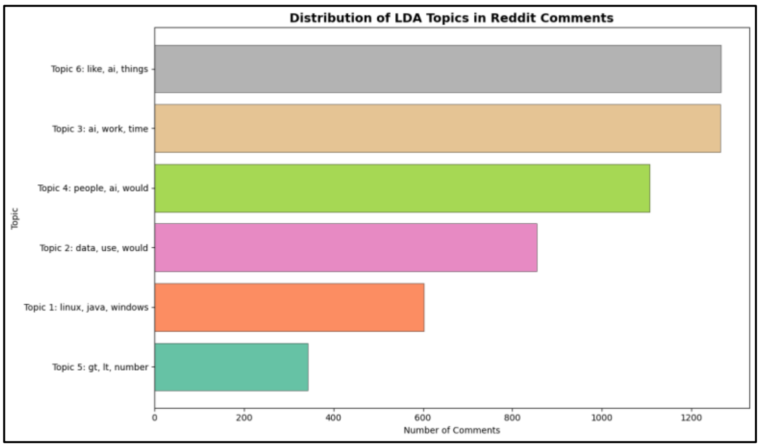

Figure 10: Distribution of LDA Topics in Reddit Comments

Figure 10 shows the distribution of comments across the six LDA topics. Topic 6 and Topic 3 were the most common, with 1,266 and 1,265 comments, reflecting broad AI, work, time, and general technology discussions. Topic 4 had 1,107 comments and the most negative average sentiment, suggesting more critical or debate-oriented discussion. Topic 1 reflected Linux, Java, and Windows themes, while Topics 2 and 5 represented data, usage, and code-like text patterns.

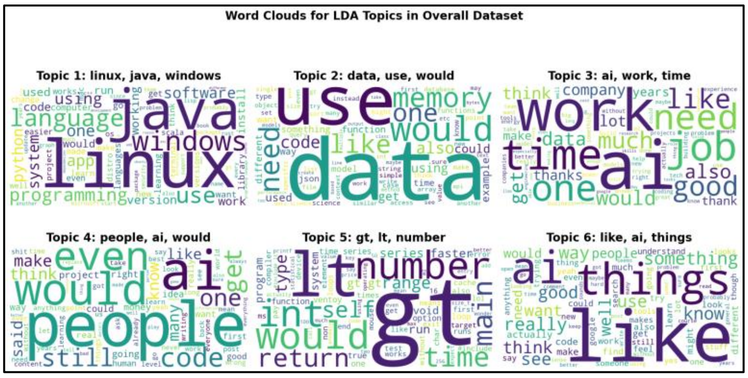

Figure 11: Word Clouds for LDA Topics in Overall Dataset

Figure 11 visually summarizes the most important words within each of the six LDA topics and supports the interpretation of Table 8 and Figure 10. Topic 1 is dominated by terms such as linux, java, windows, language, and programming, indicating operating system and programming-language discussion. Topics 3 and 6 include broader AI and discussion terms such as ai, work, job, like, and things. Topic 4 reflects more opinion-driven debate through terms such as people, ai, code, and would. Topic 5 includes code-like terms such as gt, lt, int, and return.

7.6 Summary of Sentiment and Topic Findings

RoBERTa classified most Reddit comments as neutral, reflecting the technical and informational nature of programming and technology discussions. Although VADER classified more comments as positive, RoBERTa was used as the main sentiment model because it provided a more context-aware interpretation. Sentiment varied across subreddits, although results for very small communities should be interpreted cautiously. TF-IDF confirmed the presence of key technology and programming terms such as AI, data, code, and work. LDA identified six major topics covering AI, programming languages, operating systems, data-related discussion, general technology debate, and code-like text. Section 8 links these findings with Louvain communities.

8. Linking Louvain Communities with Sentiment and LDA Topics

After analyzing the Reddit reply network and the overall NLP patterns separately, the next step was to connect the Louvain community results with sentiment and topic modeling outputs. This section examines whether the detected interaction communities also show different emotional tones and discussion themes. This integrated analysis directly addresses the fifth research question:

How do LDA topics and sentiment patterns connect with Louvain network communities?

The purpose of this section is to show that Louvain communities are not only structural groups in the reply network, but also meaningful discussion groups with identifiable sentiment and topic patterns.

8.1 Community-Level Sentiment Analysis

RoBERTa sentiment scores were grouped by Louvain community to connect sentiment patterns with network structure. Only comments from users in the reply network were included because only these users had community assignments. The RoBERTa score was calculated as positive probability minus negative probability, where values closer to +1 indicate positive sentiment, values near 0 indicate neutral or mixed sentiment, and values closer to -1 indicate negative sentiment.

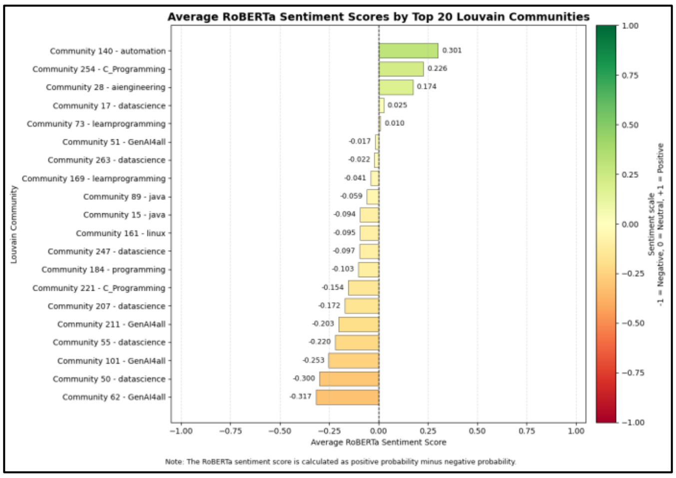

Figure 8: Average RoBERTa Sentiment Scores by Top 20 Louvain Communities

Figure 8 shows that average sentiment varied across the top Louvain communities, indicating that emotional tone was not evenly distributed across interaction groups. This is important because a single subreddit can contain multiple communities with different discussion styles. Help-based or constructive communities may show more positive sentiment, while troubleshooting, criticism, or debate-focused communities may show more negative sentiment. Therefore, Louvain communities differed not only structurally, but also emotionally.

8.2 Dominant LDA Topics in Louvain Communities

To connect topic modeling with network communities, each comment's dominant LDA topic was linked to the Louvain community of the comment author. This allowed each community to be interpreted by both its network position and its main discussion theme. The largest communities showed different topic and sentiment patterns. Community 89 was linked to the java subreddit, with 238 comments and 87 users, and its dominant topic was Topic 6: like, ai, and things, representing 34.87% of its comments. Community 51 was linked to GenAI4all, with 222 comments and 85 users, and its dominant topic was Topic 4: people, ai, and would, representing 31.08% of its comments. Overall, the results show that Louvain communities were mainly technical and discussion-oriented, with slightly negative average sentiment but mostly neutral sentiment distributions.

8.3 Word Clouds for Dominant Community Topics

To visually support the community-topic interpretation, word clouds were generated for the dominant LDA topics in the top six Louvain communities by comment volume. These word clouds help show the key vocabulary associated with each community's dominant topic.

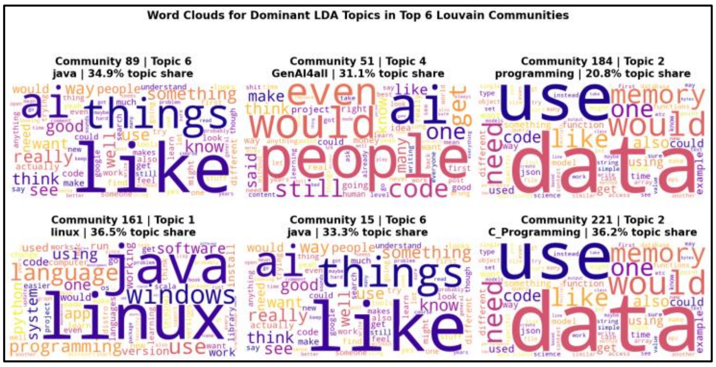

Figure 12: Word Clouds for Dominant LDA Topics in Top 6 Louvain Communities

Figure 12 shows the dominant LDA topic vocabulary for the top six Louvain communities, providing a visual explanation of their main discussion themes. Community 161 was linked to Topic 1, with terms such as linux, java, windows, and programming, supporting its interpretation as a Linux-related technical group. Community 51 was linked to Topic 4, containing terms such as people, ai, would, and code, suggesting broader AI-related debate. Community 184 was linked to Topic 2, reflecting data, memory, and code-related discussion. Overall, Figure 12 confirms that Louvain communities had identifiable textual themes.

8.4 Interpretation of Integrated Findings

The integrated analysis shows that Louvain communities were linked to distinct sentiment and topic patterns. Community detection identified users who interacted more strongly, while RoBERTa showed that these groups had different emotional tones. LDA topic modeling showed that communities also had different dominant themes. This is important because network analysis explains interaction structure, while NLP explains discussion content. Combining both methods provides a stronger interpretation of Reddit technology and programming discussions. Linux-related communities reflected operating system topics, Java communities reflected programming-language discussion, and GenAI4all communities reflected broader AI debate. Therefore, Louvain communities represent meaningful discussion groups, not only mathematical clusters.

9. Discussion and Limitations

9.1 Research Question Answers

RQ1: Which users are most central in the Reddit reply network?
Central users were identified using weighted in-degree, weighted out-degree, and PageRank. Weighted in-degree highlighted users who received repeated replies, weighted out-degree identified users who actively replied to others, and PageRank identified structurally influential users. This shows that centrality in the Reddit reply network included both attracting responses and actively participating in discussions.

RQ2: Which users act as bridges between different discussion groups?
Bridge users were identified using betweenness centrality. These users were important because they connected different parts of the network, even if they were not always the most replied-to users. This indicates that influence in Reddit discussions is not only based on popularity, but also on a user's position between different interaction groups.

RQ3: What communities are detected in the user reply network?
Louvain community detection identified 278 communities with a modularity score of 0.953, indicating strong separation between interaction groups. The largest communities were associated with subreddits such as programming, java, GenAI4all, and linux, suggesting that many interaction groups were shaped by specific technical topics or subreddit-based discussions.

RQ4: What topics and sentiments appear in Reddit technology/programming discussions?
RoBERTa sentiment analysis showed that discussions were mainly neutral, reflecting the technical and informational nature of the comments. TF-IDF and LDA topic modeling identified themes related to AI, data, programming languages, operating systems, code-like text, and broader technology discussion.

RQ5: How do LDA topics and sentiment patterns connect with Louvain network communities?
The integrated analysis showed that Louvain communities were not only structural clusters but also meaningful discussion groups with identifiable topics and sentiment patterns. For example, Linux-related communities were linked to terms such as linux, java, windows, and programming, while GenAI4all communities were linked to broader AI-related discussion. This confirms that combining network analysis with NLP provides a deeper understanding of both interaction structure and discussion content.on Answers

9.2 Limitations

This study has several limitations. The dataset was collected only from selected technology and programming-related subreddits, so findings may not represent all Reddit communities. The PullPush Reddit API may also miss comments that were deleted, removed, unavailable, or not captured during collection. The dataset was unevenly distributed, with larger subreddits such as GenAI4all, datascience, java, technology, and linux contributing more data than smaller communities such as CyberCrime and coding. This may influence sentiment and topic results. RoBERTa may still misclassify sarcasm, informal language, technical disagreement, or code-like comments. Finally, sparse-network Louvain communities may reflect short-term thread-level clusters only.

10. Conclusion

This report analyzed Reddit technology and programming comments using network analysis and NLP/text analysis. The final cleaned dataset contained 5,437 comments from 28 subreddits, and the directed weighted reply network contained 1,692 users and 1,991 reply relationships.

The network analysis identified central users, active users, bridge users, and Louvain interaction communities. The NLP analysis showed that discussions were mostly neutral in tone and covered major themes related to AI, data, programming languages, operating systems, code-like text, and broader technology discussion.

The integrated analysis demonstrated that Louvain communities were connected to identifiable topics and sentiment patterns. Therefore, the detected communities were not only mathematical network clusters, but also meaningful Reddit discussion groups. Overall, the project successfully answered the research questions by showing how users interacted, which users played important roles, what communities were detected, and what topics and sentiments appeared within those communities. Future work could collect data over a longer period, compare subreddit-specific networks, or apply advanced topic modeling methods such as BERTopic.

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Appendix

Column	Missing Values
subreddit	0
post_id	0
post_title	0
comment_id	0
author	0
body	0
parent_id	0
link_id	0
score	0
created_utc	0
created_datetime	0
Total Missing Values	**0**

Appendix Table A1: Missing Values Check

Metric	Value
Total raw comments	6,225
Unique comment IDs	6,123
Duplicate comment IDs	102
Duplicate records retained after cleaning	0

Appendix Table A2: Duplicate Comment ID Check

Invalid Record Type	Value Found in Dataset	Count
Deleted author	[deleted]	334
Removed comment body	[removed]	268
Deleted comment body	[deleted]	66

Appendix Table A3: Deleted and Removed Records Summary

Reply Target Type	Meaning	Number of Comments
Comment reply	Comment replied to another Reddit comment	3,056
Post reply	Comment replied directly to a Reddit post	2,381
Total cleaned comments		**5,437**

Appendix Table A4: Reply Target Type Distribution

Subreddit	Number of Comments	Number of Posts	Number of Authors	Average Score	Average Word Count
GenAI4all	863	40	534	1.03	32.29

datascience	772	29	465	1.06	47.85
java	502	29	256	1.83	48.69
technology	475	20	321	6.63	39.10
linux	454	18	282	1.00	40.47
C_Programming	444	41	218	1.00	58.62
learnprogramming	312	32	168	1.00	57.29
programming	246	21	155	1.00	51.93
automation	215	36	132	1.00	46.72
learnpython	161	36	70	1.04	46.11
deeplearning	133	22	88	1.00	45.91
Python	127	14	61	1.06	58.88
webdev	101	16	56	2.01	35.94
learnmachinelearning	97	24	50	1.00	49.42
aiengineering	79	32	26	1.04	39.76
OpenAI	73	14	31	1.08	40.58
dataengineering	65	19	49	1.00	68.12
algorithms	64	16	45	1.02	66.30
javascript	45	9	32	1.00	38.13
androiddev	39	12	23	1.03	34.74
generativeAI	33	22	11	1.00	217.48
MachineLearning	31	7	21	1.13	50.13
cybersecurity	26	8	19	3.42	50.31
analytics	24	8	14	1.08	89.46
dataanalysis	24	7	17	1.00	47.08
neuralnetworks	24	12	17	1.08	39.29
coding	7	4	6	1.00	31.86
CyberCrime	1	1	1	1.00	16.00

Appendix Table A5: Full Subreddit-Level Summary

Source Comment ID	**Source Author**	**Reply Target ID**	**Subreddit**	**Created Datetime**	**Target Author**
mqu9373	syscall_35	mqrvpgx	C_Programming	2025-05-06 05:44:01	EpochVanquisher
mqtisqx	nephewtom	mqsjeb4	C_Programming	2025-05-06 02:30:03	TheChief275
mqu55w2	mrheosuper	mqsjeb4	C_Programming	2025-05-06 05:09:23	TheChief275
mqumd1x	TheChief275	mqu55w2	C_Programming	2025-05-06 07:58:47	mrheosuper
mqumko4	TheChief275	mqtisqx	C_Programming	2025-05-06 08:01:04	nephewtom

Appendix Table A6: Sample User Reply Interactions

Source Author	Target Author	Edge Weight	Number of Subreddits	First Reply Time	Last Reply Time
gabieplease_	kioskinmytemporallob	10	1	2025-05-09 00:33:28	2025-05-09 02:38:49
ReallyLargeHamster	Affectionate_Cry4150	9	1	2025-05-09 05:51:42	2025-05-09 08:47:52
kioskinmytemporallob	gabieplease_	9	1	2025-05-09 00:15:31	2025-05-09 02:38:44
Affectionate_Cry4150	ReallyLargeHamster	8	1	2025-05-09 05:39:15	2025-05-09 06:44:50
jr735	activedusk	8	1	2025-05-08 22:27:04	2025-05-09 06:27:17

Appendix Table A7: Sample Weighted Edge List

	Total Comments	Number of Subreddits	Average Score	Replies Sent	Replies Received
---AI---	12	1	1.00	12	6
--TYGER--	1	1	1.00	1	0
--dany--	2	2	1.00	0	1
-MagnusBR	1	1	1.00	1	0
-Sa-Kage-	1	1	1.00	1	0

Appendix Table A8: Sample Node Attribute Table

Rank	Author	Weighted In-Degree
1	Affectionate_Cry4150	35
2	Active_Vanilla1093	23
3	bwainfweeze	16
4	ewouldblock	14
5	DannyisAbundance	14
6	Local_Ad_6109	13
7	kioskinmytemporallob	12
8	LessonStudio	11
9	orbital-state	11

10	Brilliant-Gur9384	11

Appendix Table A9: Top 10 Users by Weighted In-Degree

Rank	Author	Weighted Out-Degree
1	Minimum_Minimum4577	28
2	Affectionate_Cry4150	28
3	bwainfweeze	17
4	gece_yarisi	14
5	koflerdavid	13
6	SeaSubject9215	13
7	---AI---	12
8	FUZxxl	12
9	Local_Ad_6109	12
10	kioskinmytemporallob	11

Appendix Table A10: Top 10 Users by Weighted Out-Degree

Rank	Author	PageRank
1	Affectionate_Cry4150	0.007286
2	Desiderantes	0.005280
3	Think-Culture-4740	0.005080
4	ewouldblock	0.004986
5	cochorol	0.004932
6	AbdelMuhaymin	0.004870
7	LFCristian	0.004817
8	Tramagust	0.004813
9	TsortsAleksatr	0.004770
10	pre-medicated	0.004590

Appendix Table A11: Top 10 Users by PageRank

Rank	Author	Betweenness Centrality
1	repeating_bears	0.002224
2	hidazfx	0.002106
3	agentoutlier	0.001854
4	Hixon11	0.001465
5	__konrad	0.001440
6	jim1997jim	0.001393
7	xmsxms	0.001355
8	koflerdavid	0.001269
9	Local_Ad_6109	0.001177
10	TachosParaOsFachos	0.001099

Appendix Table A12: Top 10 Users by Betweenness Centrality

Subreddit	Number of Comments	Average Sentiment	Positive %	Neutral %	Negative %
CyberCrime	1	0.900	100.000	0.000	0.000
coding	7	0.309	42.857	42.857	14.286
analytics	24	0.286	37.500	50.000	12.500
generativeAI	33	0.281	48.485	39.394	12.121
automation	215	0.251	36.744	53.953	9.302
aiengineering	79	0.138	26.582	60.759	12.658
neuralnetworks	24	0.135	29.167	62.500	8.333
dataanalysis	24	0.094	29.167	58.333	12.500
dataengineering	65	0.064	16.923	72.308	10.769
learnmachinelearning	97	0.049	25.773	51.546	22.680
MachineLearning	31	0.000	19.355	70.968	9.677
algorithms	64	-0.019	14.062	64.062	21.875
deeplearning	133	-0.020	17.293	61.654	21.053
javascript	45	-0.034	20.000	57.778	22.222
C_Programming	444	-0.037	16.892	61.712	21.396
Python	127	-0.043	17.323	53.543	29.134
linux	454	-0.043	20.485	51.542	27.974
learnprogramming	312	-0.048	19.231	55.128	25.641
java	502	-0.051	16.733	56.972	26.295
androiddev	39	-0.059	15.385	56.410	28.205
learnpython	161	-0.068	20.497	54.658	24.845
datascience	772	-0.102	18.912	48.575	32.513
webdev	101	-0.112	18.812	50.495	30.693
OpenAI	73	-0.130	20.548	39.726	39.726
cybersecurity	26	-0.141	7.692	69.231	23.077
programming	246	-0.147	16.260	48.780	34.959
GenAI4all	863	-0.184	16.802	43.337	39.861
technology	475	-0.256	13.684	40.632	45.684

Table A13: Full RoBERTa Sentiment by Subreddit